

$x = 0$

$x = 1$



$$T(x = 0) = 5$$

$$dT/dx(x = 1) = 0$$

$$k \frac{d^2 T}{dx^2} = -1$$

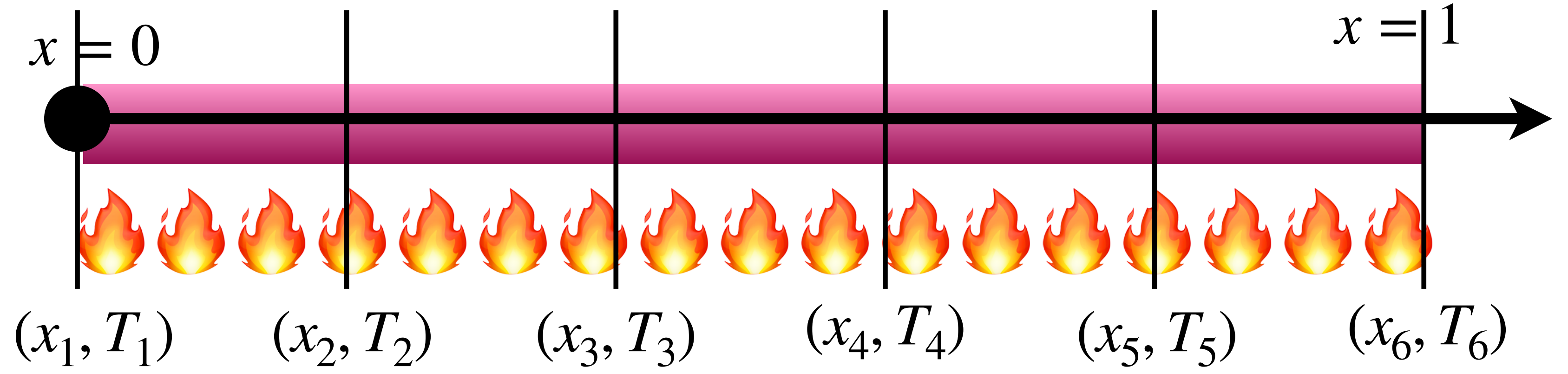
You know that $T(x = 1) = 3$, what is k ??

For 6 grid points

$$T(x = 0) = 5$$

$$k \frac{d^2 T}{dx^2} = -1$$

$$dT/dx(x = 1) = 0$$



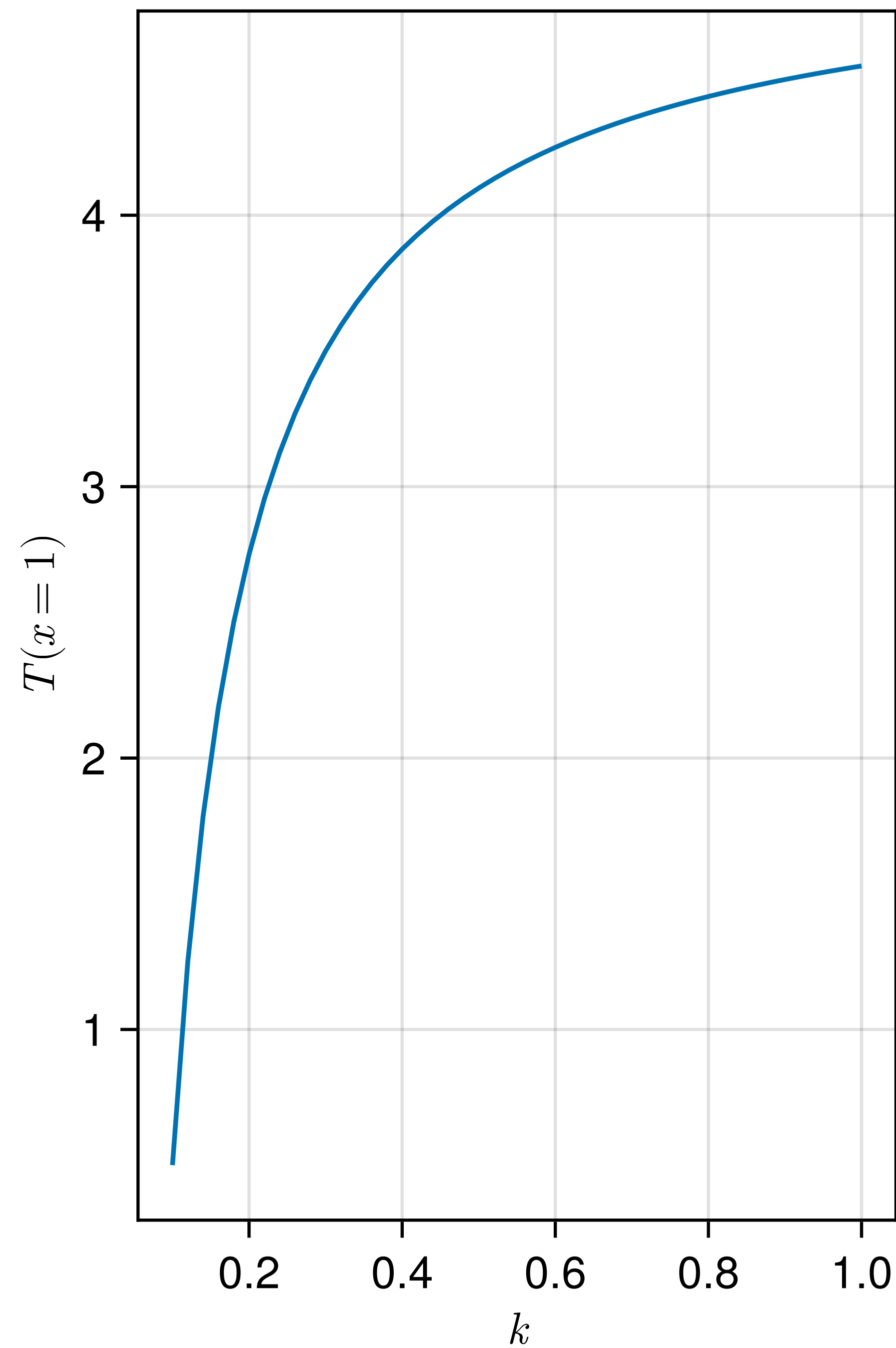
$$\begin{bmatrix} 1 & 0 & 0 & 0 & 0 & 0 \\ \frac{1}{\Delta^2} & -\frac{2}{\Delta^2} & \frac{1}{\Delta^2} & 0 & 0 & 0 \\ 0 & \frac{1}{\Delta^2} & -\frac{2}{\Delta^2} & \frac{1}{\Delta^2} & 0 & 0 \\ 0 & 0 & \frac{1}{\Delta^2} & -\frac{2}{\Delta^2} & \frac{1}{\Delta^2} & 0 \\ 0 & 0 & 0 & \frac{1}{\Delta^2} & -\frac{2}{\Delta^2} & \frac{1}{\Delta^2} \\ 0 & 0 & 0 & 0 & -1 & 1 \end{bmatrix}$$

$$\begin{Bmatrix} T_1 \\ T_2 \\ T_3 \\ T_4 \\ T_5 \\ T_6 \end{Bmatrix} = \begin{Bmatrix} 5 \\ -1/k \\ -1/k \\ -1/k \\ -1/k \\ 0 \end{Bmatrix} \quad \text{Input}$$

Output

$$\begin{bmatrix}
 1 & 0 & 0 & 0 & 0 & 0 \\
 \frac{1}{\Delta^2} & -\frac{2}{\Delta^2} & \frac{1}{\Delta^2} & 0 & 0 & 0 \\
 0 & \frac{1}{\Delta^2} & -\frac{2}{\Delta^2} & \frac{1}{\Delta^2} & 0 & 0 \\
 0 & 0 & \frac{1}{\Delta^2} & -\frac{2}{\Delta^2} & \frac{1}{\Delta^2} & 0 \\
 0 & 0 & 0 & \frac{1}{\Delta^2} & -\frac{2}{\Delta^2} & \frac{1}{\Delta^2} \\
 0 & 0 & 0 & 0 & -1 & 1
 \end{bmatrix}
 \begin{Bmatrix}
 T_1 \\
 T_2 \\
 T_3 \\
 T_4 \\
 T_5 \\
 T_6
 \end{Bmatrix}
 =
 \begin{Bmatrix}
 5 \\
 -1/k \\
 -1/k \\
 -1/k \\
 -1/k \\
 0
 \end{Bmatrix}
 \begin{matrix}
 \\
 \\
 \\
 \text{Input} \\
 \\
 \end{matrix}$$

Output



$$\begin{bmatrix}
 1 & 0 & 0 & 0 & 0 & 0 \\
 \frac{1}{\Delta^2} & -\frac{2}{\Delta^2} & \frac{1}{\Delta^2} & 0 & 0 & 0 \\
 0 & \frac{1}{\Delta^2} & -\frac{2}{\Delta^2} & \frac{1}{\Delta^2} & 0 & 0 \\
 0 & 0 & \frac{1}{\Delta^2} & -\frac{2}{\Delta^2} & \frac{1}{\Delta^2} & 0 \\
 0 & 0 & 0 & \frac{1}{\Delta^2} & -\frac{2}{\Delta^2} & \frac{1}{\Delta^2} \\
 0 & 0 & 0 & 0 & -1 & 1
 \end{bmatrix}
 \begin{Bmatrix}
 T_1 \\
 T_2 \\
 T_3 \\
 T_4 \\
 T_5 \\
 T_6
 \end{Bmatrix}
 =
 \begin{Bmatrix}
 5 \\
 -1/k \\
 -1/k \\
 -1/k \\
 -1/k \\
 0
 \end{Bmatrix}$$

Input

Output

